

Polarized neutron diffraction with *ex-situ* ^3He neutron spin filter and two-dimensional detector for a complex incommensurate magnetic order

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Polarized neutron scattering has been widely used for studying magnetic orders in condensed matter. One of the typical setups for this technique is a triple-axis spectrometer with a ferromagnetic single-crystal monochromator and analyzer, which is suited for point-by-point measurements on a horizontal scattering plane. However, this setup cannot fully meet the demands of recent studies on complex magnetic orders with emergent cross-correlated phenomena. For instance, multiferroics, magnetic skyrmions, and other topologically-nontrivial magnetic orders tend to have incommensurate magnetic modulation vectors running along various directions in reciprocal space, which cannot be captured by measurements restricted to the horizontal scattering plane. To overcome these limitations, we developed an experimental setup combining a ^3He neutron spin filter (^3He -NSF) and a two-dimensional (2D) detector, which enables us to analyze polarization of neutron scattered in out-of-plane direction. We also established a data reduction procedure to convert the 2D intensity maps into three-dimensional intensity distributions in the reciprocal space with corrections of the imperfection in ^3He polarization. This method was applied to single-crystal diffraction measurements on the multiferroic material Mn_3WO_6 , which exhibits complex incommensurate magnetic orders. We successfully observed polarization-dependent intensities of out-of-plane incommensurate magnetic reflections, demonstrating the effectiveness of the present method for analyzing complex magnetic structures.

I. INTRODUCTION

Polarized neutron scattering is a powerful technique for investigating magnetic order and excitations in condensed matter. This capability arises because neutrons interact directly with magnetic moments in the material through the magnetic dipole interaction. By analyzing the spin states of neutrons before and after scattering, magnetic scattering can be separated from nuclear scattering. The directions of magnetic moments can also be determined. To perform such polarization analysis, a neutron polarization device is required. Three techniques are commonly used for neutron polarization. The first employs Bragg reflection from a ferromagnetic single crystal, such as a Heusler crystal of Cu_2MnAl . The second uses reflection from a polarizing supermirror. The third is filtering of neutron spin state using polarized ^3He nuclei (^3He neutron spin filter, ^3He -NSF). Among these techniques, polarization analysis at triple-axis spectrometers is often performed using Heusler crystals as the neutron polarizer and analyzer [1–5], as shown in Fig. 1(a). While this type of instrument has been widely used for magnetic structure analysis, the accessible scattering vectors are for restricted to the horizontal scattering plane.

In recent years, increasing attention has been paid to

materials hosting complex magnetic structures. In such materials, the magnetic moments are often arranged in noncollinear or noncoplanar configurations, which can give rise to emergent cross-correlated phenomena. Representative examples include magnetic skyrmions and spin-hedgehog lattices [6, 7]. A common feature of these systems is that their magnetic structures are often described by multiple magnetic modulation wavevectors (q -vectors). Consequently, the associated magnetic reflections can be distributed not only within a horizontal scattering plane but throughout three-dimensional reciprocal space. To perform polarization analysis on these reflections, a large-solid-angle polarization device such as ^3He -NSF is required.

As an experimental technique for elucidating such complex magnetic structures, polarized small-angle neutron scattering (SANS) has been widely used [6, 8–10]. In polarized SANS, three-dimensional polarization analysis is already realized by combining a ^3He -NSF and a two-dimensional (2D) detector [11, 12]. In this setup, however, the detector and the ^3He -NSF are positioned directly in line with the incident beam, so the accessible region is restricted to relatively low- Q . If the same combination of ^3He -NSF and 2D detector at finite scattering angles as illustrated in Fig. 1(b), polarization analysis

can be extended to much higher- Q regions. This would make it possible to access a wide variety of magnetic reflections distributed throughout three-dimensional reciprocal space, including those associated with the complex magnetic structures described above. Related instrumental concepts have already been reported at several neutron facilities, including POLI at MLZ [13] and WOM-BAT at ANSTO [14]. Nevertheless, to the best of our knowledge, quantitative experimental demonstrations of polarization analysis using this type of setup remain limited.

In this work, we performed polarization analysis combining a ^3He -NSF and a 2D detector on the polarized neutron triple-axis spectrometer PONTA at JRR-3. Polarized neutron diffraction experiments on Mn_3WO_6 were performed using this setup. Mn_3WO_6 is a multiferroic material with a ferroaxial crystal structure belonging to the $R\bar{3}$ space group [15]. This system undergoes an antiferromagnetic phase transition at 24.5 K and exhibits spontaneous electric polarization below 22 K. In this study, polarization analysis was performed at 23 K, just below the antiferromagnetic transition. This paper mainly describes the experimental method and the data-analysis procedure. The details of the magnetic structure will be reported in a separate paper.

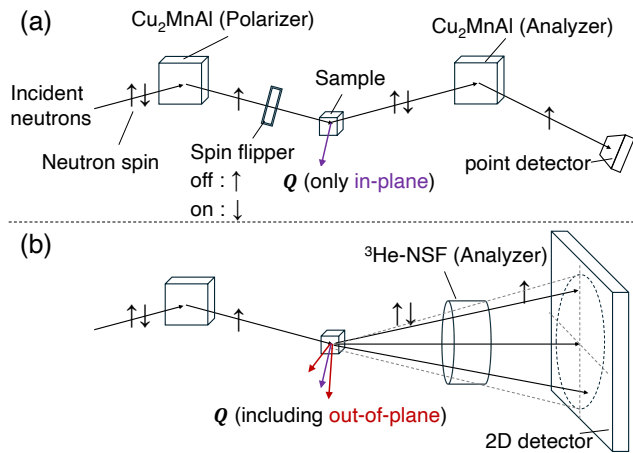


FIG. 1. Schematic illustrations comparing neutron polarization analysis techniques. (a) Conventional setup using Heusler crystals for both polarizer and analyzer, where only in-plane scattered neutrons are detected by a point detector. (b) Polarization analysis setup combining a ^3He -NSF and a 2D detector, enabling detection of both in-plane and out-of-plane scattered neutrons.

II. EXPERIMENTAL DETAILS

A. Setup

PONTA is one of the polarized neutron triple-axis spectrometers at JRR-3. It has a longitudinal polariza-

tion analysis option [5]. In this option, both the incident and scattered neutrons are polarized and analyzed by a Cu_2MnAl Heusler crystal monochromator and analyzer. In the present experiment, the Heusler analyzer and the point detector were replaced with a ^3He -NSF and a 2D detector. Fig. 2 shows an overview of the experimental setup from the sample to the detector. Scattered neutrons were detected using a scintillation-type 2D detector employing wavelength-shifting-fiber readout [16]. The active area is $256 \times 256 \text{ mm}^2$ (64 ch $\times$ 64 ch), corresponding to a channel pitch of 4 mm/ch. The detector was positioned at an angle of approximately 32° with respect to the incident neutron beam and remained fixed throughout the measurement. The sample-to-detector distance was 0.915 m. A ^3He -NSF was installed 0.43 m downstream of the sample between the sample and the detector. The cylindrical ^3He cell had a diameter of 72 mm, providing an angular acceptance of $\pm 4.0^\circ$, which covered approximately half of the central region of the detector. To reduce the background, a B_4C -shielded neutron guide tube was installed between the ^3He -NSF and the detector, and B_4C shielding was also attached around the ^3He -NSF coil. As described in the following section, polarization analysis of the scattered neutrons was performed by flipping the ^3He polarization. Therefore, the neutron spin flipper for the incident beam was not used.

A single-crystal sample of Mn_3WO_6 with a volume of 8 mm^3 was mounted on an aluminum holder and installed in a cryostat with the scattering plane set to the ($H0L$) plane. The rotation angle ω of the sample was adjustable to align the target reflection with the detector. The Helmholtz coil was used to maintain the neutron spin polarization along the neutron flight path by applying a vertical magnetic field. The magnetic field in the ^3He -NSF was oriented parallel to the beam axis, and the spins of neutrons scattered from the sample adiabatically followed the field direction, as illustrated in Fig. 2. The magnetic field strength generated by both the ^3He -NSF coil and the Helmholtz coil was approximately 1 mT.

B. ^3He neutron spin filter

The work principle of the ^3He -NSF is based on the strong spin dependence of the neutron absorption cross section of ^3He nuclei. Neutrons with spins antiparallel to the ^3He nuclear spins are strongly absorbed, while those with spins parallel to the ^3He nuclear spins are passed through. Thus, neutrons are polarized by passing through the polarized ^3He gas. The two major methods for polarizing ^3He nuclei are spin-exchange optical pumping (SEOP) [17] and metastability-exchange optical pumping (MEOP). In the present study, we employed the SEOP method. In the SEOP method, a glass cell is filled with ^3He , N_2 , and a small amount of alkali metal. The cell is irradiated with circularly polarized light resonant with an optical transition of the alkali

TABLE I. Specifications of ^3He cells.

Name	$D \times L$ (mm)	Volume (cm^3)	^3He density (amg)	N_2 pressure (Torr)	K/Rb (mol)
Kenji	72×67	140	3.2	97	26
Shimpachi	72×65	146	3.1	100	37

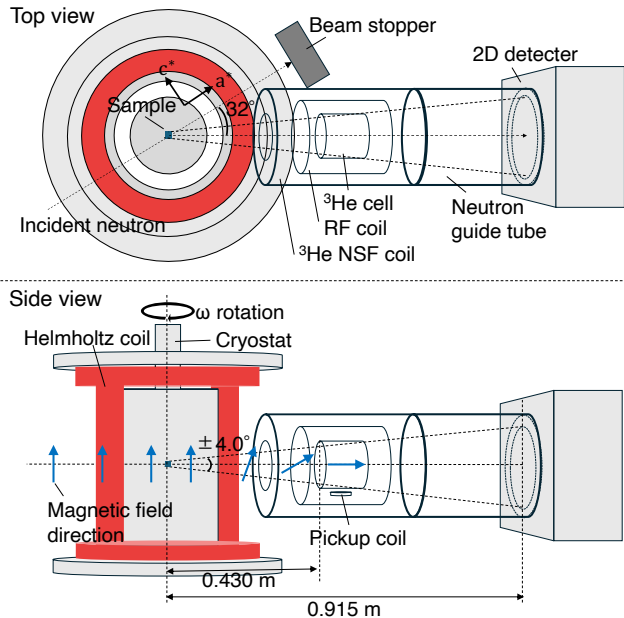


FIG. 2. Schematic illustration of the experimental setup at the 5G PONTA. Scattered neutrons from the sample pass through the ^3He -NSF coil and the neutron guide tube before reaching the 2D detector. The sample is mounted in a cryostat that can be rotated by an angle ω . The magnetic field direction induced by the Helmholtz coil is vertical, whereas the field within the ^3He -NSF is oriented along the beam axis. The neutron spins are adiabatically rotated to follow the field direction. A radiofrequency (RF) coil is used to flip the ^3He polarization, and an NMR pickup coil monitors the ^3He polarization after each spin flip.

metal. The alkali metal electrons undergo repeated excitation and de-excitation, with N_2 serving as a quenching gas to suppress radiative decay, resulting in electron spin polarization. Then, the polarization of the alkali metal is subsequently transferred to the ^3He nuclei via spin-exchange collisions, thereby achieving ^3He nuclear polarization. The SEOP system used in this study was based on the system developed for the polarized neutron spectrometer POLANO at J-PARC MLF BL23 [18]. The

SEOP system and the polarization analysis coil shown in Fig. 2 are equipped with an adiabatic fast-passage nuclear magnetic resonance (AFP-NMR) system [19]. The AFP-NMR system consists of a radio-frequency (RF) coil for flipping the ^3He polarization and an NMR pickup coil located bottom of the ^3He cell for monitoring the ^3He polarization. Two ^3He glass cells, named Kenji and Shimpachi, made of aluminosilicate glass (GE180) with nearly identical dimensions were used in this experiment. Both cells were filled with ^3He gas, nitrogen gas, and a mixture of alkali-metal atoms (Rb and K) for hybrid SEOP, which provides a higher pumping efficiency than Rb-only SEOP [20, 21]. The details of the cell specifications are described in Table I. A darkroom to house the SEOP laser system was constructed in the beamline experimental area, where the ^3He gas was polarized. After the ^3He polarization reached saturation, only the ^3He cell was transported to the ^3He -NSF coil on the beamline, where neutron measurements were performed during the relaxation of the ^3He polarization. This technique is known as the *ex-situ* ^3He -NSF method. Since the ^3He polarization decreases with time in this method, two cells were used alternately for the experiment. While one cell was used for the polarized neutron experiment, the other was polarized in the SEOP system, and the cells were exchanged every two or three days.

III. DATA REDUCTION PROCEDURE INCLUDING POLARIZATION CORRECTION

Figure 3 shows the data acquisition and reduction procedure used to obtain polarization-analysis results for magnetic reflections distributed throughout three-dimensional reciprocal space. When the sample rotation angle was fixed at $\omega = \omega_n$, we measured a pair of 2D intensity maps for the non-spin-flip and spin-flip configurations by flipping the ^3He polarization. As shown in Fig.3, we define the midpoint times of the two successive measurements as t_{2n-1} and t_{2n} , respectively. The observed scattering intensities at a detector pixel (x_i, y_j) are expressed as

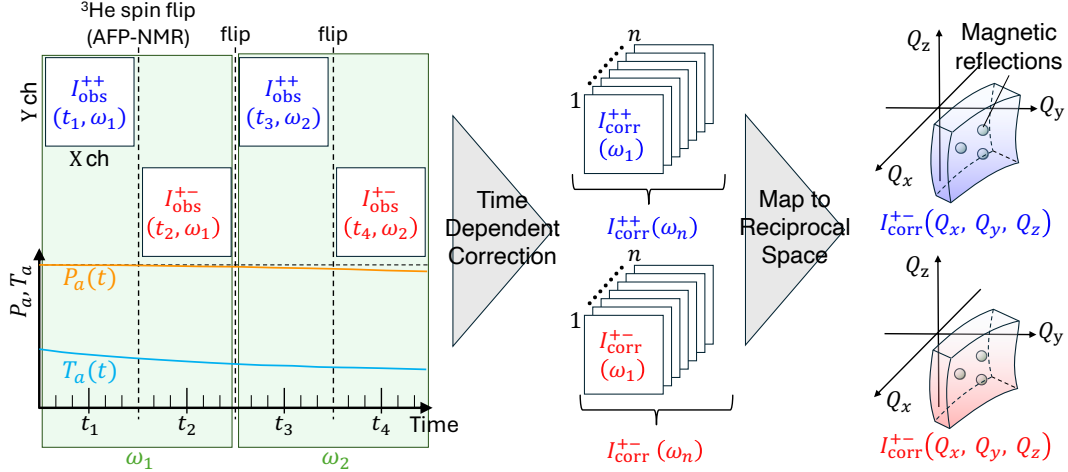


FIG. 3. Schematic diagram of the data-reduction procedure for polarization analysis using an *ex-situ* ^3He -NSF and a 2D detector. Raw 2D detector data measured at different sample rotation angles ω and times t are corrected using the time-dependent neutron polarization $P_n(t_i)$ and transmission $T_n(t_i)$. The detector coordinates (X,Y) and sample rotation angle ω are converted to Q_x, Q_y, Q_z , and then transformed to H, K, L using the UB matrix. The corrected three-dimensional reciprocal-space data can then be sliced on an arbitrary plane.

$$I_{\text{obs}}^{++}(\omega_n, t_{2n-1}, x_i, y_j) = \frac{T_p T_a(t_{2n-1})}{2} \left[(1 + P_p P_a(t_{2n-1})) I^{++}(\omega_n, x_i, y_j) + (1 - P_p P_a(t_{2n-1})) I^{+-}(\omega_n, x_i, y_j) \right] + \text{BG}(x_i, y_j), \quad (1)$$

$$I_{\text{obs}}^{+-}(\omega_n, t_{2n}, x_i, y_j) = \frac{T_p T_a(t_{2n})}{2} \left[(1 - P_p P_a(t_{2n})) I^{++}(\omega_n, x_i, y_j) + (1 + P_p P_a(t_{2n})) I^{+-}(\omega_n, x_i, y_j) \right] + \text{BG}(x_i, y_j), \quad (2)$$

where $I^{++}(\omega_n, x_i, y_j)$ and $I^{+-}(\omega_n, x_i, y_j)$ are the ideal non-spin-flip and spin-flip scattering intensities, respectively. Here, we assume an antiferromagnet in which nuclear and magnetic reflections are separated from each other. Under this condition, we can impose the conditions of $I^{++} = I^{--}$ and $I^{+-} = I^{-+}$, and simplify the equations regarding the effect of imperfection of polarization described in Appendix A. We also note that the chiral magnetic scattering, which depends on the sign of the incident neutron polarization direction, is negligible because neutron polarization and scattering vector are nearly perpendicular each other [1]. T_p and $T_a(t)$ are the neutron transmissions of the Heusler polarizer and ^3He -NSF, respectively. P_p and $P_a(t)$ are the neutron polariza-

tions of the Heusler polarizer and ^3He -NSF, respectively. $\text{BG}(x_i, y_j)$ is the background intensity at detector pixel (x_i, y_j) . In the present experiment, the background signals were mainly due to imperfect shielding of the detector and neutron flight path. Thus, we assumed that the background signals are independent of time, ^3He polarization and the sample rotation angle ω . We repeated these measurements with varying the sample rotation angle ω .

Before converting the data into the intensity distribution in the reciprocal space, we applied the polarization correction to the observed intensity maps taking into account of the relaxation of ^3He polarization in the *ex-situ* ^3He -NSF. Solving Eqs. 1 and 2 for I^{++} and I^{+-} gives

$$I^{++}(\omega_n, x_i, y_j) = \frac{1 + P_p P_a(t_{2n})}{T_p T_a(t_{2n-1}) P_p [P_a(t_{2n-1}) + P_a(t_{2n})]} [I_{\text{obs}}^{++}(\omega_n, t_{2n-1}, x_i, y_j) - \text{BG}(x_i, y_j)] - \frac{1 - P_p P_a(t_{2n-1})}{T_p T_a(t_{2n}) P_p [P_a(t_{2n-1}) + P_a(t_{2n})]} [I_{\text{obs}}^{+-}(\omega_n, t_{2n}, x_i, y_j) - \text{BG}(x_i, y_j)], \quad (3)$$

$$I^{+-}(\omega_n, x_i, y_j) = - \frac{1 - P_p P_a(t_{2n})}{T_p T_a(t_{2n-1}) P_p [P_a(t_{2n-1}) + P_a(t_{2n})]} [I_{\text{obs}}^{++}(\omega_n, t_{2n-1}, x_i, y_j) - \text{BG}(x_i, y_j)] + \frac{1 + P_p P_a(t_{2n-1})}{T_p T_a(t_{2n}) P_p [P_a(t_{2n-1}) + P_a(t_{2n})]} [I_{\text{obs}}^{+-}(\omega_n, t_{2n}, x_i, y_j) - \text{BG}(x_i, y_j)]. \quad (4)$$

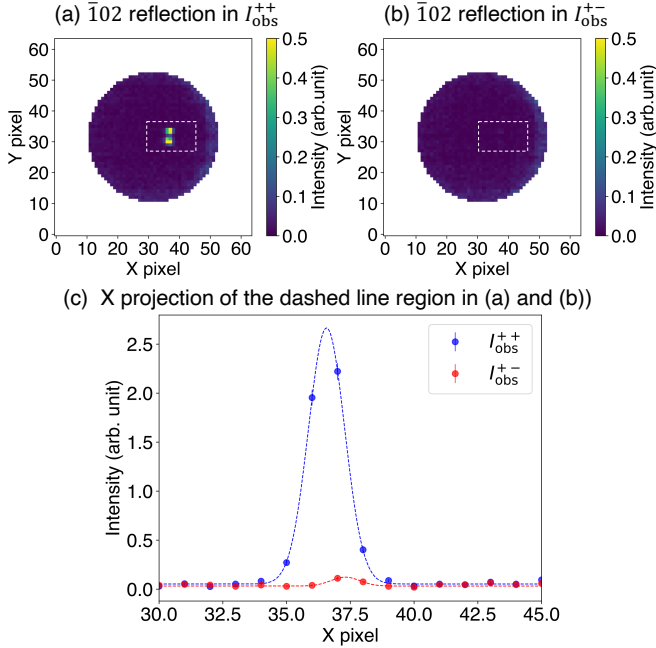


FIG. 4. (a) Observed non-spin-flip (I_{obs}^{++}), (b) spin-flip (I_{obs}^{+-}), and (c) unpolarized ($I_{\text{obs}}^{\text{unpol}}$) intensity maps, showing only the analyzing region. (d) Line profile along the x direction, obtained by integrating the intensity within the dashed regions in (a), (b), and (c).

We applied these corrections to all the pixels using the pairs of 2D intensity maps.

Finally, we convert the detector coordinates and sample rotation angle (x_i, y_j, ω_n) into reciprocal-space coordinates (Q_x, Q_y, Q_z). The resulting data are further transformed into (H, K, L) coordinates using the UB matrix, allowing the intensity distribution to be sliced along an arbitrary plane in reciprocal space.

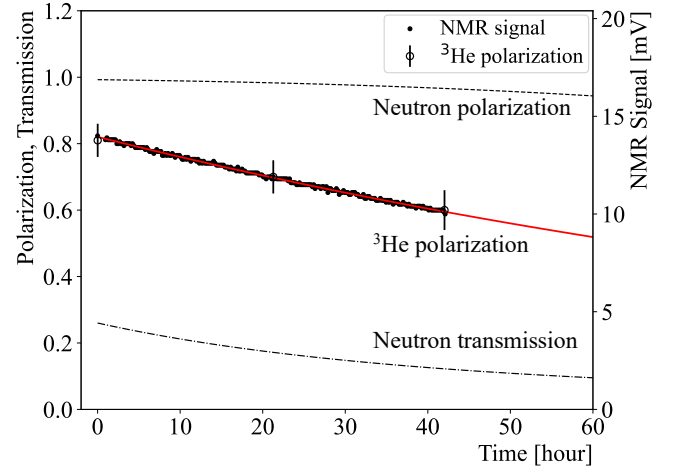


FIG. 5. Time dependence of the ^3He polarization, neutron polarization, and neutron transmission. The open circles show the ^3He polarization determined from the intensity of the 102 nuclear reflection, while the closed circles show the ^3He NMR signal. The red line represents the fitting result using an exponential decay function. The dashed and dot-dashed lines indicate the calculated neutron polarization and transmission, respectively, obtained from the ^3He polarization.

IV. RESULTS

A. Evaluation of ^3He neutron spin filter from nuclear scattering

^3He polarization is usually evaluated by measuring the transmission of a neutron beam through the ^3He cell. However, the ^3He cell cannot be placed in the direct path of the neutron beam in this experiment because of the detector's sensitivity. Therefore, the ^3He polarization was evaluated from the nuclear Bragg peak intensity. The neutron transmission $T_a(t)$, the neutron polarization $P_a(t)$, and the ^3He polarization $P_{\text{He}}(t)$ are given by the

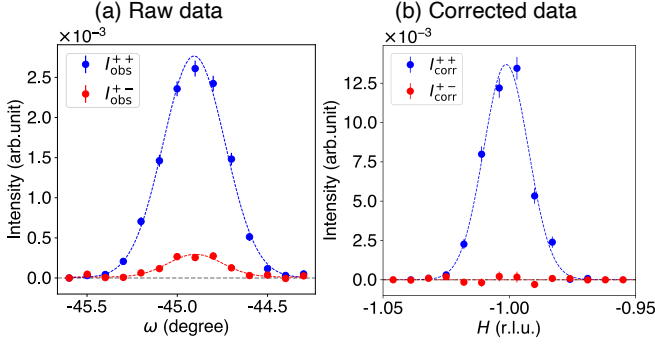


FIG. 6. The measured intensity of the $\bar{1}02$ reflection integrated over the peak region identified on the 2D detector while scanning the sample rotation angle ω . (a) Raw data, where the red and blue symbols represent the non-spin-flip intensity I_{obs}^{++} and spin-flip intensity I_{obs}^{+-} , respectively. (b) Corrected data, where the red and blue symbols represent the corrected non-spin-flip and spin-flip scattering cross sections, Σ_{obs}^{++} and Σ_{obs}^{+-} , respectively.

following equations [22]:

$$T_a(t) = T_{\text{cell}} \exp(-\mathcal{O}(\lambda)) \cosh(\mathcal{O}(\lambda) P_{\text{He}}(t)), \quad (5)$$

$$P_a(t) = \tanh(\mathcal{O}(\lambda) P_{\text{He}}(t)), \quad (6)$$

$$P_{\text{He}}(t) = P_{\text{He},0} \exp(-t/\tau), \quad (7)$$

Here, T_{cell} is the transmission of the ^3He glass cell. $\mathcal{O}$ is the opacity of ^3He cell, defined as $\mathcal{O} = \rho d \sigma$. ρ is the number density of ^3He , and d is the ^3He gas thickness. $\sigma(\lambda)$ is the neutron absorption cross section of unpolarized ^3He . $P_{\text{He},0}$ is the initial polarization at $t = 0$, and τ is the relaxation time. According to the literature, the $\sigma(\lambda)$ at $\lambda = 2.36 \text{ \AA}$ is 6992 barn [23].

First, the gas thickness ρd of each cell was determined by measuring a nuclear Bragg reflection. In the present measurement, we selected the $\bar{1}02$ reflection which has a similar scattering angle to that of the magnetic reflections of interest. Before polarizing ^3He by SEOP, we measured the nuclear peak at ω which satisfy the Bragg condition through the unpolarized ^3He cell. From the intensity map, we extracted a box-integrated intensity near the peak. By subtracting the background signals estimated from the intensities in the surrounding the pixels, we obtained the integrated intensity of the Bragg reflection, $\mathcal{I}_{\text{obs}}^{\text{unpol}}$. By using eqs 1, 5, 7 and $P_{\text{He}} = 0$, $\mathcal{I}_{\text{obs}}^{\text{unpol}}$ is expressed as follows.

$$\mathcal{I}_{\text{obs}}^{\text{unpol}} = \sum_{(i,j) \in R} \left(\frac{T_p}{2} T_{\text{cell}} \exp[-\mathcal{O}(\lambda)] I^{++}(x_i, y_j) \right). \quad (8)$$

where R denotes the box-integrated region. Note that the nuclear Bragg reflection contains only the non-spin-flip scattering intensity, and thus $I^{+-} = 0$. The same measurement was performed using an empty cell with the same dimensions as the ^3He glass cell to obtain $\mathcal{I}_{\text{obs}}^{\text{empty}}$, where the transmission term $T_{\text{cell}} \exp[-\mathcal{O}(\lambda)]$ is replaced

by T_{cell} . Then, taking the ratio $\mathcal{I}_{\text{obs}}^{\text{unpol}} / \mathcal{I}_{\text{obs}}^{\text{empty}}$ allows the gas thickness ρd to be determined. As a result, the ρd values of the Kenji and Shimpachi cells were determined to be $18.5 \pm 0.5 \text{ amg cm}$ and $18.8 \pm 0.4 \text{ amg cm}$, respectively.

Second, the $P_{\text{He}}(t)$ was determined. After polarizing the ^3He by SEOP, the ^3He cell was transported to the beamline, and then the I_{obs}^{++} and I_{obs}^{+-} of the nuclear reflection were measured. As shown in Fig. 4, we clearly observed the nuclear Bragg peak in the non-spin flip channel. Note that the peak in Fig. 4(a) shows the tiny splitting because of the mosaicity of the single crystal sample. Here, we defined the time origin ($t = 0$) as the center of the measurement times of I_{obs}^{++} and I_{obs}^{+-} . Since these measurement times was much shorter than the ^3He relaxation time, both intensities were assumed to be measured at the same ^3He polarization. From Eqs. 1, 2, and 5, the ^3He polarization $P_{\text{He}}(t)$ at $t = 0$ ($P_{\text{He},0}$) satisfies following relationship:

$$\frac{\mathcal{I}_{\text{obs}}^{++}(t=0) + \mathcal{I}_{\text{obs}}^{+-}(t=0)}{\mathcal{I}_{\text{obs}}^{\text{unpol}}} = 2 \cosh(\mathcal{O}(\lambda) P_{\text{He},0}). \quad (9)$$

When deriving this equation, we exploited the fact that the scattering cross section of the nuclear reflection contains only the non-spin-flip scattering component. Using the measured intensities in Eq. 9, the initial ^3He polarization of the Shimpachi cell was determined to be $P_{\text{He},0} = 0.81 \pm 0.05$.

The decay of the ^3He polarization was determined from the NMR signal of the ^3He -NSF. Throughout the experiment, the NMR signal was recorded during each magnetic scattering measurement at every ω angle. Figure 5 shows the time evolution of the ^3He polarization of Shimpachi cell determined from the nuclear reflection (open circles) and the NMR signal (closed circles). The relaxation time τ was estimated by fitting the NMR signal with Eq. 9, yielding $\tau = 131.1 \pm 0.7$ hours. We also measured nuclear scattering intensities several times during the magnetic scattering measurements. The time evolution of the nuclear reflection intensities and the NMR signal showed good agreement. As a result, the time dependence of the ^3He polarization $P_{\text{He}}(t)$ was determined using these $P_{\text{He},0}$ and τ values, respectively. The same analysis was performed for each ^3He cell.

The polarization of the Heusler polarizer, P_p , was also determined from the same nuclear reflection data. From Eqs. 1 and 2, the asymmetry between I_{obs}^{++} and I_{obs}^{+-} is given by

$$\frac{I_{\text{obs}}^{++} - I_{\text{obs}}^{+-}}{I_{\text{obs}}^{++} + I_{\text{obs}}^{+-}} = P_p P_a(t). \quad (10)$$

Since $P_a(t)$ is independently determined from the ^3He polarization, P_p can be obtained from this relation. The average value of P_p was found to be 0.86 ± 0.06 . Note that P_p includes the effects of environmental background and neutron depolarization in the guide field.

The validity of the polarization correction (Eq. 4) was confirmed using the ω scan of the $\bar{1}02$ reflection. At each ω , the intensity was integrated within the peak region identified on the 2D detector (Fig. 4). The raw intensity profiles of the $\bar{1}02$ reflection show a finite spin-flip intensity owing to the imperfect neutron polarization, as shown in Fig. 6(a). By applying the polarization correction using $P_a(t)$, $T_a(t)$, and P_p , the corrected non-spin-flip and spin-flip intensities, I_{corr}^{++} and I_{corr}^{+-} , were obtained, as shown in Fig. 6(b). Since only the non-spin-flip scattering cross section Σ^{++} is expected for nuclear scattering, the result indicates that the polarization correction was successfully performed. Details of the polarization correction are described in Appendix A. These values were used for the polarization correction of the magnetic scattering data.

B. Polarization analysis of magnetic scattering

This subsection presents the polarization-analysis results for the magnetic reflections of Mn_3WO_6 at antiferromagnetic phase. First, intensity maps of antiferromagnetic reflections were measured using unpolarized neutrons and a 2D detector. The scattering plane was set to the $(H0L)$ plane, a conventional triple-axis setup would mainly provide a reciprocal-space map in this plane, as shown in Fig. 7(d). However, the 2D detector enables data analysis along the out-of-plane direction, as demonstrated in Fig. 7(b). Here, we can observe the three magnetic reflections $\mathbf{q}_1 = (0.059, 0.019, 0.5)$, and $\mathbf{q}_2$ and $\mathbf{q}_3$ correspond to threefold rotations of $\mathbf{q}_1$ about the c^* axis. These $\mathbf{q}$ values show that $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ cannot lie within a single 2D plane passing through the reciprocal-space origin (Fig. 7(a)). This highlights why three-dimensional reciprocal-space measurements are essential for capturing such magnetic structures. Figures 7(c) and 7(e) show the polarization-analysis maps obtained from the intensity difference $I_{\text{corr}}^{++} - I_{\text{corr}}^{+-}$. Clear positive and negative contrasts are observed at the magnetic reflection positions identified in the unpolarized intensity maps. In addition, to compare the spin-dependent intensities more quantitatively, line profiles along the H direction through $\mathbf{q}_3$ and $\mathbf{q}_1$ are shown in Figs. 7(f) and 7(g), respectively. The relative intensities of the non-spin-flip and spin-flip components differ clearly between the two reflections. These polarization-analysis data provide constraints on the magnetic moment components associated with each propagation vector. A detailed magnetic structure analysis based on these constraints will be presented in a separate paper. These results indicate that the spin-dependent scattering intensities were successfully resolved for magnetic reflections distributed in three-dimensional reciprocal space.

V. SUMMARY AND OUTLOOK

In this work, we developed a polarization analysis method that combines ^3He -NSF with a 2D detector to analyze arbitrary reciprocal-lattice points distributed in three-dimensional reciprocal space. We demonstrated a correction procedure for the time-dependent neutron polarization and transmission using a nuclear Bragg reflection and confirmed its validity. Polarization analysis was successfully performed for the incommensurate magnetic reflections of Mn_3WO_6 at antiferromagnetic phase, and the details of the magnetic structure will be reported in a separate paper. The positions of the ^3He -NSF and the 2D detector were fixed in the present experiment, whereas a configuration in which both devices are mounted on a scanning arm is planned to cover a wider region of reciprocal space. Although the present *ex-situ* method requires time-dependent correction, *in-situ* method unnecessary the time-dependent correction because SEOP is performed continuously throughout the measurement on the beam path. Recently, a compact *in-situ* ^3He -NSF system has been developed at J-PARC, which can be mounted on a scanning detector arm[24]. This implementation is expected to maintain high neutron polarization and transmission throughout the experiment, thereby facilitating polarization analysis of complex magnetic structures over a wider region of reciprocal space and for a broader range of materials.

ACKNOWLEDGMENTS

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Appendix A: Polarization correction with *ex-situ* ^3He neutron spin filter

In this section, the polarization correction in the present experimental setup is described. This correction is based on the polarization matrix reported by A. Wildes [25, 26]. A spin-dependent measured count for the non-spin flip configuration (I_{obs}^{++}) is expressed as follows:

$$\begin{aligned} I_{\text{obs}}^{++} = & N_0 T_p p_p \Sigma^{++} T_a p_a \\ & + N_0 T_p p_p \Sigma^{+-} T_a (1 - p_a) \\ & + N_0 T_p (1 - p_p) \Sigma^{-+} T_a p_a \\ & + N_0 T_p (1 - p_p) \Sigma^{--} T_a (1 - p_a), \end{aligned} \quad (\text{A1})$$

Here, N_0 is the total number of incident neutrons, assumed to include equal populations of the (+) and (−) spin states. T_p and T_a are the total neutron transmissions of the polarizer and analyzer, respectively. p_p and p_a are the polarization efficiencies of the polarizer and analyzer,

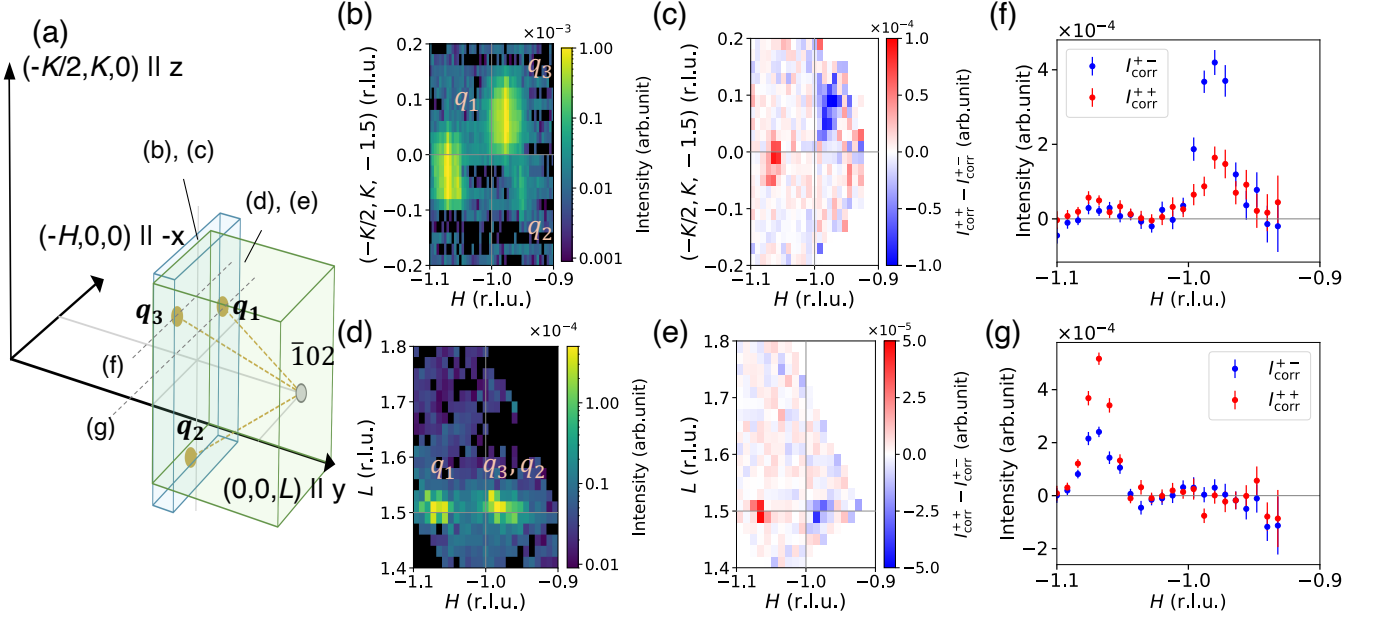


FIG. 7. (a) Schematic illustration of the reciprocal-space region measured in the present experiment. The translucent volumes labeled (b), (c) and (d), (e) indicate the regions integrated along the L direction and the $(-K/2, K, 0)$ direction, respectively. The dashed lines along the $-H$ direction through q_3 and q_1 indicate the line cuts shown in panels (f) and (g), respectively. (b), (d) Unpolarized neutron intensity maps in the $(H, K/2, -1.5)$ and $(H, 0, L)$ planes, respectively. (c), (e) Polarization-analysis maps obtained from the difference between the corrected non-spin-flip and corrected spin-flip intensities, $I_{\text{corr}}^{++} - I_{\text{corr}}^{+-}$. (f), (g) Line profiles of the (I_{corr}^{++}) and the (I_{corr}^{+-}) at q_3 and q_1 , respectively.

respectively. Σ^{++} , Σ^{+-} , Σ^{-+} , and Σ^{--} are the scattering cross sections for the corresponding spin states. The derivation of this equation is schematically illustrated in Fig. 8. The polarization efficiencies p_p and p_a are defined within the ranges $0.5 \leq p_p \leq 1$ and $0.5 \leq p_a \leq 1$. In the present experimental setup, only the I^{++} and the I^{+-} configurations can be measured. The chiral contribution and the nuclear-magnetic interference term are assumed to be negligible, such that the scattering cross sections satisfy $\Sigma^{++} = \Sigma^{--}$ and $\Sigma^{+-} = \Sigma^{-+}$. Under these assumptions, Eq. A1 can be rewritten into the following matrix form that includes the spin configuration:

tion:

$$\begin{bmatrix} I_{\text{obs}}^{++} \\ I_{\text{obs}}^{+-} \end{bmatrix} = N_0 T_p T_a \begin{bmatrix} p_p & 1 - p_p \\ 1 - p_p & p_p \end{bmatrix} \begin{bmatrix} p_a & 1 - p_a \\ 1 - p_a & p_a \end{bmatrix} \begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix}. \quad (\text{A2})$$

Since the *ex-situ* ^3He -NSF exhibits a decay of the ^3He polarization over time, both the neutron polarization efficiency p_a and the neutron transmission T_a become time-dependent. Accordingly, by treating Eq. A2 as a time-dependent matrix equation and taking its inverse, the following expression is obtained:

$$\begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix} = \frac{1}{N_0 T_p P_p (P_a(t_1) + P_a(t_2))} \begin{bmatrix} 1 + P_p P_a(t_2) & -1 + P_p P_a(t_1) \\ -1 + P_p P_a(t_2) & 1 + P_p P_a(t_1) \end{bmatrix} \begin{bmatrix} 1/T_a(t_1) & 0 \\ 0 & 1/T_a(t_2) \end{bmatrix} \begin{bmatrix} I_{\text{obs}}^{++}(t_1) \\ I_{\text{obs}}^{+-}(t_2) \end{bmatrix}, \quad (\text{A3})$$

where p_p and p_a are converted to P_p and P_a using the relations $P_p = 2p_p - 1$ and $P_a = 2p_a - 1$. $T_a(t)$, P_p , and $P_a(t)$ are experimentally determined and described

in Sec. IV A. In the main text, N_0 and T_p are treated as a scale factor and are absorbed into the corrected intensities $I_{\text{corr}}^{\pm\pm}$.

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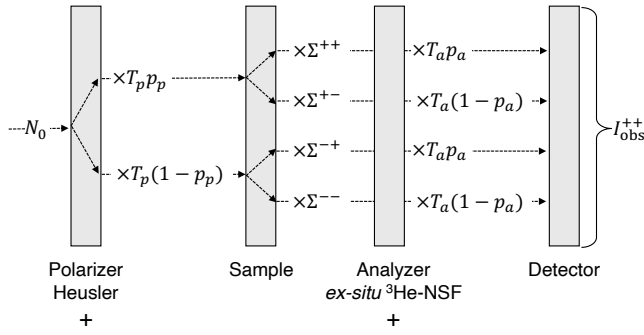


FIG. 8. Schematic diagram of the spin-dependent neutron intensity in the present experimental setup. The measured intensity I^{++} is given by the sum of the contributions from the four spin-scattering channels when the Heusler crystal and the ^3He -NSF are both polarized in the same direction.

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